

Do EA Sports FC 25 Player Ratings Reflect Real Market Value?

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Introduction

Football video games such as EA Sports FC 25 assign numerical ratings to players with the aim of capturing their real on-pitch performance and quality. In

parallel, platforms like Transfermarkt estimate players' market values based on performance, age, league, reputation and transfer activity.

I focus on two questions mainly:

1. How strongly do EA FC 25 ratings correlate with players' market values in the 2024 season?
2. How do age, position and player type affect this relationship, and which players are undervalued or overvalued relative to their EA-based profiles?

To answer these questions, I built a merged dataset combining EA FC 25 ratings and Transfermarkt valuations, conduct descriptive analysis and statistical tests, and then use linear regression and k-means clustering to model and interpret the link between ratings and value.

Data

I used two public Kaggle datasets:

- EA Sports FC 25 Dataset: For players' overall ratings and attributes. (<https://www.kaggle.com/datasets/nyagami/ea-sports-fc-25-database-ratings-and-stats/data>)
- Transfermarkt Dataset: For players' real market values. (https://www.kaggle.com/datasets/davidcariboo/player-scores/data?select=player_valuations.csv)

To prepare the transfermarkt data:

- Filtered all valuations to 2024 in transfermarkt data that contains valuations and by doing that we also eliminated retired players too.
- Then merged this data using player_id with another transfermarkt data which contains information such as name, position, sub-position, date of birth, country and current club.
- Also computed the age in 2024 and log of market value to reduce skewness

To merge the EA FC 25 data with Transfermarkt data:

- I create a variable which is name_clean in both datasets by converting names to lower case and stripping spaces.
- Also eliminated the duplicates.
- Then merged the datasets based on name_clean variable.
- At the end to I created a data that contains only important columns which are "player_id,name,name_clean,OVR,PAC,SHO,PAS,DRI,DEF,PHY,Position,Age,

Nation,League,Team,position,sub_position,age_2024,country_of_citizenship,current_club_name,league_2024,current_club_domestic_competition_id,market_value_2024_eur,log_value_2024" and called it "merged_core".

Methods

1. Exploratory Data Analysis (EDA)

I showed the distribution of ratings and market values, and visualize patterns by position and age using:

- Histograms of "OVR" and "log market value".
- Correlation between **OVR** and `log_value_2024`.
- Heatmaps:
 - Position × Market value bins
 - Age × Market value bins
- Hypothesis tests:
 - Correlation test: is the OVR–value correlation statistically significant?
 - t-test: high-rated players ($OVR \geq 85$) vs low-rated players ($OVR < 80$).
 - ANOVA across positions and age groups.

2. Machine learning models

Supervised Learning:

- Modeled `log_value_2024` as Y
- For X: used EA attributes, age, positions and leagues
- Split the data into train and test sets, fit a "Linear Regression" model, and evaluate it with RMSE, MAE and R^2 on the test set.
- Refitted the model on the full dataset, computed predicted values and `value_gap_eur = market_value_2024_eur - pred_value_eur`
- Used this to identify the 10 most undervalued players.

Unsupervised Learning:

- I applied "k-means clustering (k = 4)"
- Features: `OVR`, `PAC`, `SHO`, `PAS`, `DRI`, `DEF`, `PHY`, `age_2024`
- For each cluster, I computed "Number of players", "Average ratings", "average age", "average market_value_2024_eur"
- This gives us player types which are "High OVR, prime age attacking players with the highest average market value", "Young, lower OVR but relatively pacey players with modest all-round stats and low average market value", "Prime age defensive players (high DEF and PHY) with

midrange market values", "Average OVR, defensively oriented players with very low attacking stats and relatively low market values".

Discussion

Overall, the analysis shows that EA FC 25 ratings are meaningfully connected to real market values. The strong correlation between OVR and log market value, the large and significant difference between high- and low-rated players, and the performance of the regression model all point in the same direction: EA ratings contain substantial information about what the market is willing to pay.

However, the relationship is not one-dimensional. Age, position and player type play important roles:

- Age: Market values tend to be higher in “prime years” and lower for older players, even if their ratings remain decent.
- Position: Attacking and creative roles are typically more expensive than purely defensive ones, especially in top leagues.
- Player type: The clustering results highlight that prime-age attacking profiles form the most expensive group, while young prospects and defensively oriented players with lower attacking stats remain relatively cheap.

The supervised model also allows us to identify individual players whose market values are far below their predicted values. These cases can be discussed as potentially undervalued players according to the EA-based model, though real-world factors (contracts, injuries, club situations) must also be considered.

Limitations and Conclusion

This project has several limitations:

- EA ratings may lag behind players’ current performance and do not update continuously.
- Transfermarkt values are estimates, not actual transfer fees, and may contain subjective elements.
- Matching players by cleaned names can introduce matching errors, especially for common names.
- The linear regression model is relatively simple and may not capture all nonlinear interactions between attributes.

Despite these limitations, our findings support the idea that EA FC 25 ratings reflect real-world market values to a significant extent, but with important systematic differences across age, position and player profile.

References

Cariboo, D. *Football data from Transfermarkt* [Data set]. Kaggle. Retrieved January from https://www.kaggle.com/datasets/davidcariboo/player-scores/data?select=player_valuations.csv (kaggle.com)

Nyagami, D. *EA SPORTS FC 25 database, ratings and stats* [Data set]. Kaggle. Retrieved from <https://www.kaggle.com/datasets/nyagami/ea-sports-fc-25-database-ratings-and-stats/data> (kaggle.com)